

rest of the lab was on a virtual holiday. He was not able to take leaves like his other colleagues as he had joined the company recently. He started working on his idea to bring all the parts of the chip under one block of the semiconductor, as one monolithic unit. The result—a slice of a centimetre-wide germanium, with protruding wires, glued to glass slide.

He gathered several executives, including former Texas Instruments Chairman Mark Shepherd, for a demonstration event on September 12, 1958. When Kilby pressed the switch, an unending sine curve undulated across the oscilloscope screen. We have the first IC, a 'phase-shift oscillator.' The patent for the first IC, 'Solid Circuit made of Germanium,' was filed on February 6, 1959, and the world never looked back.

There is an unprecedented growth in the ICs and microprocessors, but did Jack Kilby take all the credit for this? No! He said, "Well, I don't know that I get credit for their profound effect. It's true that the original idea was mine, but what you see today is the work of probably tens of thousands of the world's best engineers, all concentrating on improving the product, reducing the cost, things of that sort." Kilby was very right when further improvements and developments are the handiwork of a number of engineers and scientists. Let us begin from the beginning. Let us look at those who "have had even a small part in helping turn the potential of human creativity into practical reality."

Transistor: the starting point

Early morning on November 1, 1956, William Shockley received a telephone call informing him that he had won the Nobel Prize in physics along with John Bardeen and Walter Brattain for the invention of the transistor. Nine years ago on December 23, 1947, they invented the point contact transistor at Bell Laboratories in Murray Hill, New Jersey. The name 'transistor' was coined by John R. Pierce.

The first silicon transistor was presented by Morris Tanenbaum at Bell

Labs on January 26, 1954. Gordon Teal, with expertise in high-purity crystals, takes the credit for the first commercial silicon transistor in 1954.

Shockley subsequently designed a junction transistor. He was well known for his smart examples. Once a student confessed his inability to understand the concept of amplification. Shockley told him, "Take a bale of hay and tie it to the tail of a mule. Then strike a match and set the bale of hay on fire. Now compare the energy expended shortly thereafter by the mule with the energy expended in striking the match, you will understand the concept of amplification."

Shockley left Bell Labs and in September 1955 founded the Shockley Semiconductor Laboratory. He recruited "the most creative team in the world for developing and producing transistors," which included Gordon Moore, Jean Hoerni and Robert Noyce.

In 1949, Professor Grant Gale at Grinnell College showed his 18 physics students two of the very first transistors ever made from Bell Labs. Noyce was one of them and he was immediately hooked to the transistor. When he joined Massachusetts Institute of Technology for his Ph.D., he knew more about transistors than most of his professors.

Soon afterwards, Noyce joined Philco Corporation which was not ready to invest money into the futuristic research Noyce had in mind. In 1956, he left Philco to join Shockley. The way he joined was a classic example of his confidence. He contacted Shockley by telephone a few times and put himself and his wife on a night flight from Philadelphia to San Francisco. They arrived in Palo Alto at 6 am, and by noon Noyce had signed a contract to buy a house. Then he met Shockley and got his job, in that order.

But by December 1956 their egos clashed and most of that 'most creative team' got disenchanted with Shockley's management style. In the summer of 1957 Moore, Hoerni, Jay Last and four other engineers wanted to look for greener pastures by starting their own

company. But they needed a leader and an administrator. So they turned to Noyce. He was 29 years old. "With his strong face, his athlete's build and the Gary Cooper manner, Bob Noyce projected what psychologists call the halo effect. People with the halo effect seem to know exactly what they are doing and, moreover, make you want to admire them for it. They make you see the halos over their heads." He agreed to join them but with his white lab coat and goggles on and his research in. They founded Fairchild Semiconductor.

Jay Last said in an interview much later, "There were eight of us. We all had different skills but in the group we had all the necessary skills and it was a completely cooperative effort." Shockley called them 'Traitorous Eight.'

Everyone knows that the first electronic numerical integrator and computer known as ENIAC was a monster measuring 30 metres long and 3 metres high, which boasted use of 18,000 vacuum tubes. But the government wanted smaller computers to facilitate automatic on-board guidance in rockets. Transistors did simplify the system and could cut down the size. But then even a radio with seven or eight transistors looked like a map of a small city and had to be hand wired in a cumbersome, laborious process. Sizes were getting reduced and smaller devices were being produced. 'Miniature' was no longer the word and the new buzz word was 'micro-miniature.'

Fairchild's founders understood that it is the survival of the micros. Noyce and Moore theorised an idea of combining transistors in a solid block of silicon. Transistors, insulators, rectifiers, resistors, capacitors and all of them would have to be carved, etched and built on a wafer of silicon or, in other words, an entire circuit to be fabricated on a little chip.

However, in the late 1958, Kurt Lehovec, at the Sprague Electric Company, found a simple solution to the isolation problem. He was paid only one dollar for this invention by the